by a shovel or spade till the mix is of uniform in colour. Then water is added and the whole mass is mixed for ten to fifteen minutes.

4.2.5.2 Machine Mixing



In machine mixing, cement and sand are just mixed in a concrete mixer and then water is added gradually. Mixing is carried out for more than one minute till the mixture is brought to a plastic condition.

Mortars within two hours can be used after retempering by adding water to restore the consistency. After two hours the mix should not be used.

ACTIVITY 3

Visit a construction site near by your school and collect details about mixing of mortar.

4.2.6 Uses of Mortar

1. To bind the building units such as bricks, stones, etc.
2. To carry pointing and plaster work on exposed surface of masonry.
3. To form an even bedding layer for building units.
4. To form joints of pipes.
5. To improve the appearance of structure.
6. To prepare moulds.
7. To serve as a matrix to hold coarse aggregates, etc.

4.2.7 Precautions to be taken in Using Mortar

The following precautions are to be taken while making use of mortar.

i) After preparation, it should be consumed as early as possible.

ii) The cement mortar should be consumed before its initial setting time starts.

iii) It is advisable to prepare mortar of one bag of cement at a time.

iv) It is advisable to stop the work in frosty weather.

v) The building units should be soaked in water before mortar is applied.

vi) The construction work should be cured by sprinkling water to avoid rapid drying.

vii) The mortar should not contain excess water and it should be as stiff as it can be conveniently used.

4.2.8 Required Strength of Mortars In Masonry

Strength of mortars to be used for joining bricks, stones, blocks, etc., should depend on the strength of the materials. There is no advantage in using over strong mortar. It should be sufficiently strong to resist erosion, abrasion and other factors affecting durability. The following list gives the ratio of common cement – sand mortars used in practice in Tamilnadu.

1. Damp proof course - CM 1 : 2
2. General Brick work - CM 1 : 6
3. Stone masonry – CM 1:6
4. Arch work - CM 1:3
5. Pointing work- CM 1:1 to 1:3
6. Brick work plaster - CM 1:5
7. RC Plasterwork (like ceiling) - CM 1:3 to 1:4

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